

D911

Transforming Healthcare Through Technology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 40-provider clinic that recently added telehealth services encrypts all laptops and sets automatic logoff after 10 minutes of inactivity. The IT director previously worked at a bank. Under the HIPAA Security Rule, these are which type of safeguard?

- A. Physical safeguards, because laptops are physical devices
- B. Banking-style controls outside HIPAA
- C. Administrative safeguards
- D. Technical safeguards

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A hospital performs an annual security risk analysis and requires security training for its workforce. Which type of HIPAA safeguard are these?

- A. None; training is never required
- B. Physical safeguards
- C. Administrative safeguards
- D. Technical safeguards

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A ransomware attack at a regional hospital known for its burn unit encrypts the EHR, and clinicians cannot open records for three days. Investigators find no evidence that data were copied out. The hospital recently hired a new security chief. Which part of the CIA triad was primarily affected?

- A. Confidentiality
- B. Integrity
- C. All three equally, since every attack hits all three
- D. Availability

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A nurse finds that a patient's allergy list was changed without authorization, removing a penicillin allergy. The record was still accessible and not disclosed. Which part of the CIA triad was affected?

- A.** Confidentiality
- B.** Integrity
- C.** Availability
- D.** None, because nothing was stolen

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A home-health agency that recently expanded to three counties discovers on March 1 that an unencrypted laptop holding records of 2,000 patients was stolen. The laptop was a year old. What does the HIPAA Breach Notification Rule require?

- A.** Notify affected individuals, HHS, and prominent local media within 60 days of discovery
- B.** Log it and report to HHS once a year
- C.** Notify affected individuals only
- D.** No notice, since the laptop was stolen rather than hacked

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A clinic discovers a breach of unsecured PHI affecting 120 patients. What notification timing applies?

- A.** No notification, since breaches under 500 never need reporting
- B.** Individuals, HHS, and media all within 60 days of discovery
- C.** Notify local media within 60 days
- D.** Individuals within 60 days of discovery; HHS within 60 days after the end of the calendar year

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A health system with a popular patient portal wants a third-party diabetes app to pull patients' lab results from its EHR through a modern API. The system's CIO recently spoke at a national conference. Which standard is built for this?

- A. HL7 FHIR
- B. Any standard, since all standards exchange data the same way
- C. SNOMED CT
- D. ICD-10-CM

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

After a hospital that just earned a top safety grade adds 40 new drug-interaction alerts, clinicians override 95% of alerts, including critical ones. The alerts came preconfigured from a large national vendor. What is the best response?

- A. Remove or quiet low-value alerts and tier the rest by severity
- B. Make every alert a hard stop
- C. Switch to a smaller vendor
- D. Add more alerts so clinicians see more warnings

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A hospital with a new innovation lab buys an AI tool that screens photos for skin cancer. The tool was trained mostly on images of light-skinned patients. Its developer won an industry award. The hospital serves a racially diverse population. What is the main risk?

- A. No risk, because AI decisions are objective
- B. The main risk is that it runs too slowly
- C. Lower accuracy for patients with darker skin, so the tool must be validated locally across groups
- D. No risk, because the developer won an award

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

Before deploying a sepsis-prediction model, which step best checks for bias?

- A.** Rely on the accuracy the vendor reports
- B.** Check only overall accuracy
- C.** Test the model's performance separately for each age, sex, and racial group in local data
- D.** Remove the race variable, which always eliminates bias

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A health network that recently merged with a rival wants to use blockchain. The board chair became enthusiastic after reading technology news. Which use best fits blockchain's strengths?

- A.** Storing full patient records on a public blockchain for privacy
- B.** A shared ledger that tracks drug shipments or provider credentials across partner organizations
- C.** Replacing all encryption in the EHR
- D.** Any use, since blockchain makes all data private automatically

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

At a hospital with a new outpatient pavilion, an employee enters her password on a fake login page from a phishing email. She has worked there for 15 years. Which control would best stop an attacker from logging in with the stolen password?

- A.** Nothing, since a stolen password always gives full access
- B.** Multifactor authentication
- C.** Antivirus software on her workstation
- D.** A longer password requirement

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A 40-provider clinic ~~that recently added telehealth services~~ encrypts all laptops and sets automatic logoff after 10 minutes of inactivity. ~~The IT director previously worked at a bank.~~ Under the HIPAA Security Rule, these are which type of safeguard?

Target word: safeguard type **Decides:** Controls built into the technology.

A	Confusion set	Physical safeguards protect facilities and equipment, such as locks.
B	Camouflage	The director's background is irrelevant; these are HIPAA technical safeguards.
C	Adjacent category	Administrative safeguards are policies, risk analysis, and training.
D	Correct	Encryption and automatic logoff are technical controls.

2. Answer C Concept 1 • transfer

A hospital performs an annual security risk analysis and requires security training for its workforce. Which type of HIPAA safeguard are these?

Target word: safeguard type **Decides:** Policies and training.

A	Absolute language	Security awareness training is a required administrative standard.
B	Adjacent category	Physical safeguards protect buildings and devices.
C	Correct	Risk analysis and training are administrative.
D	Confusion set	Technical safeguards are software and system controls.

3. Answer D Concept 2

A ransomware attack at a regional hospital ~~known for its burn unit~~ encrypts the EHR, and clinicians cannot open records for three days. Investigators find no evidence that data were copied out. ~~The hospital recently hired a new security chief.~~ Which part of the CIA triad was primarily affected?

Target word: CIA element **Decides:** Records could not be accessed.

A	Confusion set	No data were shown to have been exposed.
B	Adjacent category	Integrity concerns unauthorized changes to data content.
C	Absolute language	Attacks can affect one element more than others.
D	Correct	Clinicians could not access records when needed.

4. Answer B Concept 2 • transfer

A nurse finds that a patient's allergy list was changed without authorization, removing a penicillin allergy. The record was still accessible and not disclosed. Which part of the CIA triad was affected?

Target word: CIA element **Decides:** Data were altered.

A	Adjacent category	Nothing was disclosed to unauthorized people.
B	Correct	Unauthorized changes to data harm integrity.
C	Confusion set	The record could still be opened.
D	Dropped element	Drops the unauthorized change, which is an integrity problem.

5. Answer A Concept 3

A home-health agency ~~that recently expanded to three counties~~ discovers on March 1 that an unencrypted laptop holding records of 2,000 patients was stolen. ~~The laptop was a year old.~~ What does the HIPAA Breach Notification Rule require?

Target word: required notification **Decides:** 500 or more affected.

A	Correct	Large breaches require all three notices within 60 days.
B	Confusion set	The annual log applies to breaches under 500.
C	Dropped element	Drops the HHS and media notices required at 500 or more.
D	Adjacent category	How the data were lost does not matter; unencrypted PHI was compromised.

6. Answer D Concept 3 · transfer

A clinic discovers a breach of unsecured PHI affecting 120 patients. What notification timing applies?

Target word: timing for a small breach **Decides:** Fewer than 500 affected.

A	Absolute language	Individuals and HHS must still be notified.
B	Confusion set	That is the rule for 500 or more.
C	Wrong stem number	Media notice is triggered at 500 residents of a state, not 120.
D	Correct	Small breaches go to individuals promptly and to HHS in an annual report.

7. Answer A Concept 4

A health system ~~with a popular patient portal~~ wants a third-party diabetes app to pull patients' lab results from its EHR through a modern API. ~~The system's CIO recently spoke at a national conference.~~ Which standard is built for this?

Target word: exchange standard **Decides:** API-based data exchange.

A	Correct	FHIR defines API-based exchange of health data.
B	Absolute language	Code sets and exchange standards do different jobs.
C	Confusion set	SNOMED is a clinical vocabulary, not an exchange method.
D	Adjacent category	ICD-10 codes diagnoses; it does not move data between systems.

8. Answer A Concept 5

After a hospital ~~that just earned a top safety grade~~ adds 40 new drug-interaction alerts, clinicians override 95% of alerts, including critical ones. ~~The alerts came preconfigured from a large national vendor.~~ What is the best response?

Target word: best response **Decides:** Alert fatigue from too many low-value alerts.

A	Correct	Fewer, more relevant alerts reduce overrides of critical ones.
B	Absolute language	Hard stops on everything block care and cause workarounds.
C	Camouflage	The vendor's size is not the problem; the alert design is.
D	Inverted operation	More alerts worsen alert fatigue.

9. Answer C Concept 6

A hospital ~~with a new innovation lab~~ buys an AI tool that screens photos for skin cancer. The tool was trained mostly on images of light-skinned patients. ~~Its developer won an industry award.~~ The hospital serves a racially diverse population. What is the main risk?

Target word: main risk **Decides:** Training data do not match the patients.

A	Absolute language	Models inherit the limits of their data.
B	Adjacent category	Speed is not the issue the training data raise.
C	Correct	Unrepresentative training data cause biased performance.
D	Camouflage	An award does not show the tool works for this population.

10. Answer C Concept 6 · transfer

Before deploying a sepsis-prediction model, which step best checks for bias?

Target word: bias check **Decides:** Subgroup performance on local data.

A	Confusion set	Vendor results may come from a different population.
B	Dropped element	Overall accuracy can hide poor performance in a subgroup.
C	Correct	Bias shows up as different performance across subgroups.
D	Absolute language	Other variables can stand in for race; removal does not guarantee fairness.

11. Answer B Concept 7

A health network ~~that recently merged with a rival~~ wants to use blockchain. ~~The board chair became enthusiastic after reading technology news.~~ Which use best fits blockchain's strengths?

Target word: best-fit use **Decides:** A shared, tamper-evident record across organizations.

A	Inverted operation	Public, permanent ledgers work against privacy and correction.
B	Correct	Multi-party, tamper-evident tracking is blockchain's strength.
C	Adjacent category	Blockchain is not a substitute for encryption.
D	Absolute language	Blockchain is not automatically private.

12. Answer B Concept 8

At a hospital ~~with a new outpatient pavilion~~, an employee enters her password on a fake login page from a phishing email. ~~She has worked there for 15 years.~~ Which control would best stop an attacker from logging in with the stolen password?

Target word: best control **Decides:** A second factor beyond the password.

A	Absolute language	MFA exists precisely to block this.
B	Correct	A password alone is not enough to log in when MFA is required.
C	Adjacent category	Antivirus does not stop logins with stolen credentials.
D	Dropped element	A longer password is still stolen when typed into a fake page.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	2	
Absolute language	Always, never, guarantee — where judgment applies.	9	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	HIPAA Security Rule safeguards	Administrative safeguards are policies, risk analysis, and training; physical safeguards protect facilities and devices; technical safeguards are controls in the technology such as encryption, access control, audit logs, and automatic logoff.	Calling encryption physical because it runs on a laptop.	Whether the control is a policy, a lock, or a software setting.
2	The CIA triad	Confidentiality keeps data from unauthorized eyes, integrity keeps it accurate and unaltered, and availability keeps it accessible when needed.	Assuming every attack is a confidentiality breach.	Records that can't be opened, were changed, or were exposed.
3	Breach notification	For breaches of 500 or more, notify individuals, HHS, and prominent media within 60 days of discovery; smaller breaches go to individuals within 60 days and to HHS in an annual log.	Thinking notification depends on how the data were lost or only goes to patients.	A lost or stolen device with a patient count.
4	Interoperability standards	HL7 FHIR is the modern API-based standard for exchanging health data; ICD-10 and SNOMED CT are code sets that describe content.	Treating a coding vocabulary as an exchange standard.	An app that needs to pull data from an EHR.
5	Clinical decision support and alert fatigue	CDS works when alerts are few, relevant, and tiered by severity; too many low-value alerts drive overrides of critical ones.	Adding more alerts or hard stops to fix override rates.	A high override rate after alerts were added.
6	AI bias and validation	AI models reflect their training data; validate performance across subgroups in the local population before and after deployment.	Believing AI is objective or that removing one variable removes bias.	A model trained on a population unlike the patients it will serve.
7	What blockchain is good for	Blockchain is a shared, tamper-evident ledger suited to multi-party records like supply chains, credentials, and audit trails, not to storing PHI or replacing encryption.	Thinking blockchain makes data private automatically.	A proposal to put health data "on the blockchain."
8	Multifactor authentication	MFA requires a second factor, so a stolen password alone cannot open the account; it is a key defense against phishing.	Thinking longer passwords or antivirus stop use of a phished password.	An employee who entered a password on a fake login page.

Course scope (public description): Transforming Healthcare Through Technology explores how information technology (IT) is utilized in healthcare settings to improve patient care, operational efficiency, and decision-making processes. The course includes the integration of IT systems with healthcare management principles to support organizational goals and regulatory requirements (compliance with HIPAA). Topics also include electronic health records (EHR), mobile applications, and digital health technologies. Healthcare cybersecurity, including threats and vulnerabilities specific to healthcare IT systems, will be emphasized. Learners also explore emerging technologies such as artificial intelligence (AI), machine learning, blockchain, and their applications in healthcare. There are no prerequisites to the course.